

Grade 6 Accelerated
Unit 13
Lesson 1 Homework

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1. What is the value of the expression $7 \cdot 4 - (x + 4)$ if $x = 2$?

2. Find the value of $8 - 3(4 - y) + 2y \cdot 6$ when $y = 4$.

3. Evaluate $3p + \frac{6}{j} - 1$ for $p = 3$ and $j = 8$.

4. Evaluate $10 - 20 \div 2f + 11$ if $f = 9$.

5. Find the value of $7 \cdot (m - 3m + 2)$ when $m = 1$.

6. Find the value of $7 - (y + 12)^2 + 6m$ if $m = 4$ and $y = 3$.
7. Find the value of $r^3 \div x \cdot c$ if $r = 4$, $x = 8$ and $c = 3$.
8. Evaluate the expression $3 \cdot (u^2 - 4) + \frac{k}{2}$ for $u = 2$ and $k = 6$.
9. What is the value of the expression $5g \cdot h^3 + (h - 1)^2$ if $g = 2$ and $h = 5$?
10. What is the value of the expression $\left(\frac{b^2-6}{4}\right) \cdot 7 + y^2$ if $b = 4$ and $y = 7$?

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Lesson 2 Homework

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1. Evaluate the expression $8b + 10 - (9 - 6)^2$ for $b = -3$.

2. Evaluate $-7m - 5q^2 + \frac{s}{2}$ when $m = 8$, $q = -2$, and $s = -48$.

3. Evaluate $2s - 6q - f^2$ for $s = 4$, $q = -6$, and $f = -5$.

4. Evaluate $-m^3 - 3d + (n - y)$ if $m = 3$, $d = 7$, $n = 9$, and $y = 1$.

5. Evaluate $4g^4 - \frac{z}{4}$ when $g = 2$ and $z = 16$.

6. What is the value of $5(8x + 4) - 3m^2$ if $x = -3$ and $m = 5$?
7. Find the value of $-9(h - 6) + 2a - (9 - a)^5$ if $h = 8$, and $a = 7$.
8. Evaluate the expression $p^3 - 5k(y - k)$ for $p = -4$, $k = 6$ and $y = 10$.
9. Evaluate the expression $8 \div j - 12 \div (-10 + 4)$ if $j = 2$.
10. What is the value of the expression $4m - 10 + (a - 11) \cdot 4^3$ if $m = 9$ and $a = 5$?

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Lesson 3 Homework

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Determine if each pair of expressions are equivalent.

1. $-9(20 + 7)$ and $-9(20) - 9(7)$

2. $4.5(10 + 7)$ and $4.5 \cdot 10 + 4.5 \cdot 7$

3. $\frac{3}{4} + \frac{9}{10}$ and $\frac{9}{10} + \frac{3}{4}$

4. $\frac{2}{3} - \frac{1}{2}$ and $\frac{1}{2} - \frac{2}{3}$

5. $\frac{1}{6} \times \frac{2}{9}$ and $\frac{2}{9} \times \frac{1}{6}$

6. $\frac{7}{11} \div \frac{1}{3}$ and $\frac{1}{3} \div \frac{7}{11}$

7. $\frac{5}{7} - \left(\frac{1}{2} - \frac{3}{4}\right)$ and $\left(\frac{5}{7} - \frac{1}{2}\right) - \frac{3}{4}$

8. $\left(\frac{1}{5} \div \frac{2}{5}\right) \div \frac{8}{9}$ and $\frac{1}{5} \div \left(\frac{2}{5} \div \frac{8}{9}\right)$

9. Rewrite $(y - 11) + 26$ using the associative property.

10. Rewrite $z + 14$ using the commutative property.

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Lesson 4 Homework

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1. Rewrite the expression $(2z + 6) + 9z$ using the associative property.

2. Rewrite the expression $(2z + 6) + 9z$ using the commutative property.

3. Rewrite the expression $5 \cdot (-3 \cdot 4g)$ using the associative property.

4. Rewrite the expression $5 \cdot (-3 \cdot 4g)$ using the commutative property.

5. Rewrite the expression $-22r + 2(8k + 2)$ using the distributive property.

6. Rewrite the expression $-22r + 2(8k + 2)$ using the commutative property of multiplication.
7. Rewrite the expression $-22r + 2(8k + 2)$ using the commutative property of addition.

Write two equivalent algebraic expressions for each expression.

8. $(5 + 7) \cdot 3$

9. $4t \cdot 3$

10. $-5x + 2 - 10$

Lesson 5 Homework

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Write an algebraic expression for each written description.

5. The difference of twenty-seven and a number.

6. Gina will give each classmate, c , 5 free carnival ride tickets.
7. There are 4 soda cases, each with s , number of cans in each case.
8. A bag of candy has 250 calories per serving, s .
9. A YouTube video pays \$0.01 per click, c , to the creator of the video.
10. Frank scored 16 points less on the quiz than his classmate Jeannine, j .

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Write an algebraic expression to represent each real-world situation.

1. Olivia has 15 videogames downloaded to her device. Her friend has x more videogames than her. How many songs does her friend have?

2. Trevor has x tickets from a local arcade. His brother has 15 times more tickets than Trevor. How many tickets does his brother have?

3. James earned 60 points while playing a card game. Arthur earned x fewer points than James. How many points did Arthur earn?

4. Javier has 28 books and wants to put an equal number of books on x bookshelves. How many books go on each shelf?

5. Francesca has a plant that measured h inches tall last week. This week, the plant is 12 inches tall.

For each expression, create a real-world scenario that the algebraic expression could represent.

6. $x - 5$

7. $18x$

8. $\frac{150}{y}$

9. $19 + k$

10. $62 - h$

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Lesson 7 Homework

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Write an equivalent expression by combining like terms.

1. $3x + 6x$

2. $-4y + 8 + y - 5$

3. $-y + -y + y + y$

4. $7y - 2 + 6 - 2y$

5. $10 + y - 9 + y$

Find the sum of each expression.

6. $2 + 9x - 1 + 5x =$

7. $4x + 1 + 10x - 8 =$

8. $-3y + 1 + 2 + 6y =$

9. $10 + 2y - 8 + y =$

10. $6y - 8 + y + 3 =$

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Find the sum of each expression using the fewest terms possible.

1. $(0.3x + 0.6) + (0.7x - 0.8)$

2. $\left(-\frac{4}{5} + 8y\right) + \left(5y - \frac{1}{5}\right)$

3. $(-0.5y - 2.4) + (y + 7.3)$

4. $\left(7\frac{1}{2}y - \frac{2}{5}\right) + \left(6\frac{3}{5} - 2\frac{1}{4}y\right)$

5. $\left(10 - \frac{1}{3}x\right) + \left(-9 + \frac{1}{3}x\right)$

Determine the value of each expression.

6. $(5.2 + 9.3x) + (-1.8x + 8.5) =$

7. $\left(\frac{3}{8}x + \frac{1}{4}\right) + \left(\frac{3}{4}x - \frac{5}{16}\right) =$

8. $(-3.9y + 5) + (3.9 + 6y) =$

9. $\left(1.0 + \frac{2}{9}y\right) + (-8.6 + x) =$

10. $\left(10.8 + \frac{2}{7}x\right) + \left(-\frac{9}{7} + x\right) =$

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Lesson 9 Homework

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Find the difference of each expression using the fewest terms possible.

1. $(0.4x + 0.9) - (0.5x - 0.7)$

2. $\left(-\frac{3}{5} + 7y\right) - \left(4y - \frac{2}{5}\right)$

3. $(-0.4z + 1.9) - (z + 6.2)$

4. $\left(5\frac{1}{2}y - \frac{3}{5}\right) - \left(4\frac{1}{5} - 1\frac{3}{4}y\right)$

5. $\left(10 - \frac{2}{9}q\right) - \left(-9 - \frac{2}{9}q\right)$

Simplify each of the expressions.

6. $(8.2 - 5.6x) - (1.4x - 3.5) - (2.9x - 7.5)$

7. $\left(\frac{3}{8}x - \frac{1}{4}\right) + \left(\frac{1}{4}x + \frac{11}{16}\right) + \left(\frac{5}{8}x - \frac{3}{4}\right)$

8. $(3.9y + 7) - (3.9 + 4y) + (-3.9 + 4y)$

9. $\left(1.8 + \frac{2}{9}r\right) + (-8.6 + r) - \left(-5.6 + \frac{1}{9}r\right)$

10. $\left(\frac{1}{6} + \frac{2}{7}n\right) - \left(-\frac{5}{6} + n\right) - \left(3\frac{1}{6} + 4\frac{4}{7}n\right)$

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Lesson 10 Homework

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Determine if the expressions are equivalent for the given values.

	Value	$7x + 3x - 6$	$2(5x - 8) + 10$	Equivalent?
1.	$x = -3$			☐ yes ☐ no
2.	$x = 0$			☐ yes ☐ no
3.	$x = 0.3$			☐ yes ☐ no
4.	$x = \frac{1}{2}$			☐ yes ☐ no

5. Depending on the value being substituted into the expressions, the evaluated

expressions

- ☐ always
☐ sometimes
☐ never

had the same value as each other.

Determine whether the expressions are equivalent for the given values.

		$\frac{1}{3}(9 + 6y) + 4x$	$2\left(2x + y + \frac{3}{2}\right)$	Equivalent?
6.	$x = -2$ $y = 6$			☐ yes ☐ no
7.	$x = 0$ $y = 1$			☐ yes ☐ no
8.	$x = \underline{\hspace{1cm}}$ $y = \underline{\hspace{1cm}}$			☐ yes ☐ no

Determine if the following algebraic expressions are equivalent using any method you choose.

	Expression A	Expression B	Equivalent?
9.	$x(9 - 11) - \frac{3}{5}(10 + 15x)$	$-11x - 15$	☐ yes ☐ no
10.	$\frac{5}{9}(6x + 12x) - (25 + 3x)$	$-25 + 10x$	☐ yes ☐ no

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Lesson 11 Homework

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From the specified set of integers, determine which, if any, make the equation true.

$\{-4, -1, 1, 8\}$

1. $12 + 5w = 6w + 4$

2. $3 - x = 2x + 6$

3. $3y - 5 = 8y + 15$

4. $2u + 6 = 9u - 1$

Determine if the inequality $18 \geq 2m - 6$ is true for each value.

5. 8

6. 12

7. 15

Determine if the inequality $-y + 3 + 8y \geq -3y - 7$ is true for each value.

8. -5

9. 0

10. 5

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Lesson 12 Homework

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Write an equivalent expression using the associative, commutative, or distributive property. State the property used.

1. $7x - \frac{3}{7}(14x - 21)$

2. $(16x + 13) - 2x + 5$

3. $\left(-\frac{2}{3}y + 8\right) \cdot 9$

4. $-8\left(\frac{3}{4}x + 2\right) + (4 - 2x)$

5. If expressions are equivalent, then they will

- ☐ always
- ☐ sometimes
- ☐ never

have the same terms when they are simplified using properties of operations.

Use properties of operations to determine if the expressions are equivalent.

6. $0.7x - 0.1(4x - 2)$ and $7x - (4x - 2)$

7. $(5x + 12) - 6$ and $7x - 2(x - 3)$

8. $2\frac{1}{5}x - \frac{3}{10} - x$ and $\frac{6}{5}\left(x - \frac{2}{5}\right)$

9. $0.2 + 0.1(14x - 2)$ and $2x - \left(\frac{3}{5}x - \frac{1}{5}\right)$

10. Expressions that are equal are also equivalent.

- ☐ True
- ☐ False